EEUC201 • ANALOG AND DIGITAL ELECTRONICS

Combinational Logic Circuits

Adders • Subtraction • ALU • Comparators • Encoders/Decoders • MUX/DEMUX

Course Module Presentation

ROADMAP

What We Will Cover

1. Combinational Logic — Overview
2. Half Adder
3. Full Adder & Ripple Carry
4. Binary Subtraction
5. Arithmetic Logic Unit (ALU)
6. Digital Comparators
7. Priority Encoder
8. Decoders
9. Multiplexer
10. Demultiplexer
11. Summary

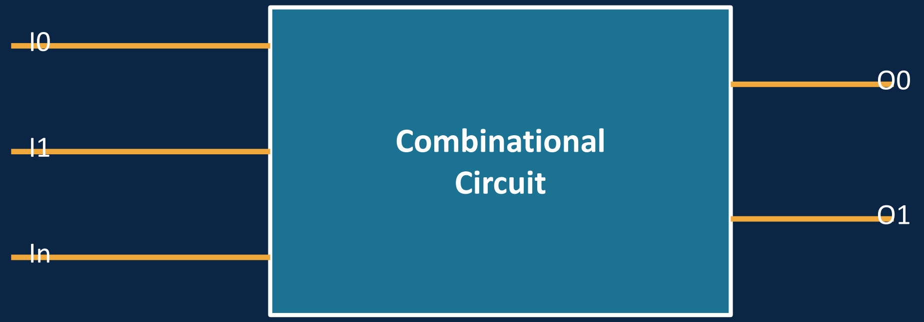
FOUNDATIONS

What is Combinational Logic?

A combinational circuit is a digital circuit whose output at any instant depends only on the present combination of inputs — it has no memory of past inputs.

- Built purely from logic gates (AND, OR, NOT, XOR, etc.)
- No feedback path and no clock/memory elements
- Output = f(current inputs) only
- Described using Boolean expressions, truth tables, or K-maps

BLOCK DIAGRAM



n inputs → m outputs, related purely by Boolean logic — no internal state.

- Examples: adders, comparators, encoders, decoders, MUX/DEMUX

Half Adder

Adds two single-bit binary numbers, A and B, producing a Sum and a Carry — but cannot accept a carry-in from a previous stage.

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

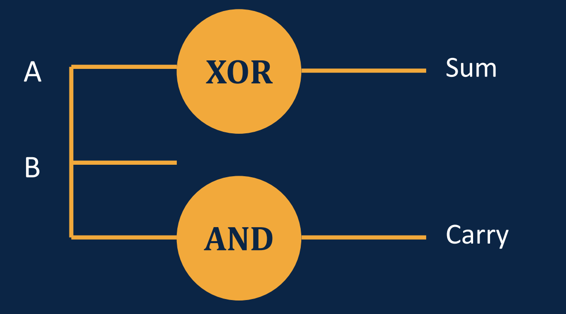
EQUATIONS

$$\text{Sum} = A \oplus B$$

$$\text{Carry} = A \cdot B$$

Sum → XOR gate
Carry → AND gate

GATE-LEVEL CIRCUIT



LIMITATION

A half adder has only 2 inputs, so it cannot add a carry generated from a lower-order bit position. To build multi-bit adders, we need a circuit that also accepts a carry-in — the Full Adder.

Full Adder

Adds three bits — A, B, and a Carry-in (Cin) — producing a Sum and a Carry-out (Cout). Building block of multi-bit binary adders.

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

n Full Adders chained → n-bit Ripple Carry Adder

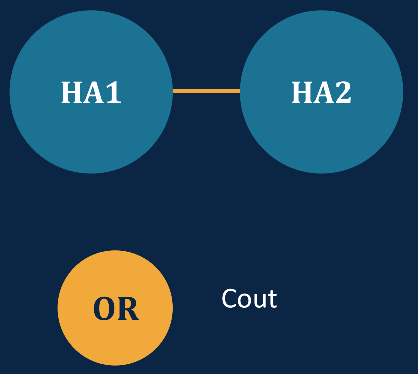
EQUATIONS

Sum =
$$A \oplus B \oplus Cin$$

Cout =
$$AB + BCin + ACin$$

Cout can also be written as $AB + Cin(A \oplus B)$

BUILT FROM 2 HALF ADDERS



A, B → HA1 → partial sum & carry
Partial sum, Cin → HA2 → Sum
Carries OR'd → Cout

Binary Subtraction

Half Subtractor

Inputs: A, B

Subtracts B from A for a single bit (no borrow-in).

$$\text{Diff} = A \oplus B$$

$$\text{Borrow} = A' \cdot B$$

Full Subtractor

Inputs: A, B, Bin

Subtracts B and a Borrow-in from A, producing Diff and Borrow-out.

$$\text{Diff} = A \oplus B \oplus \text{Bin}$$

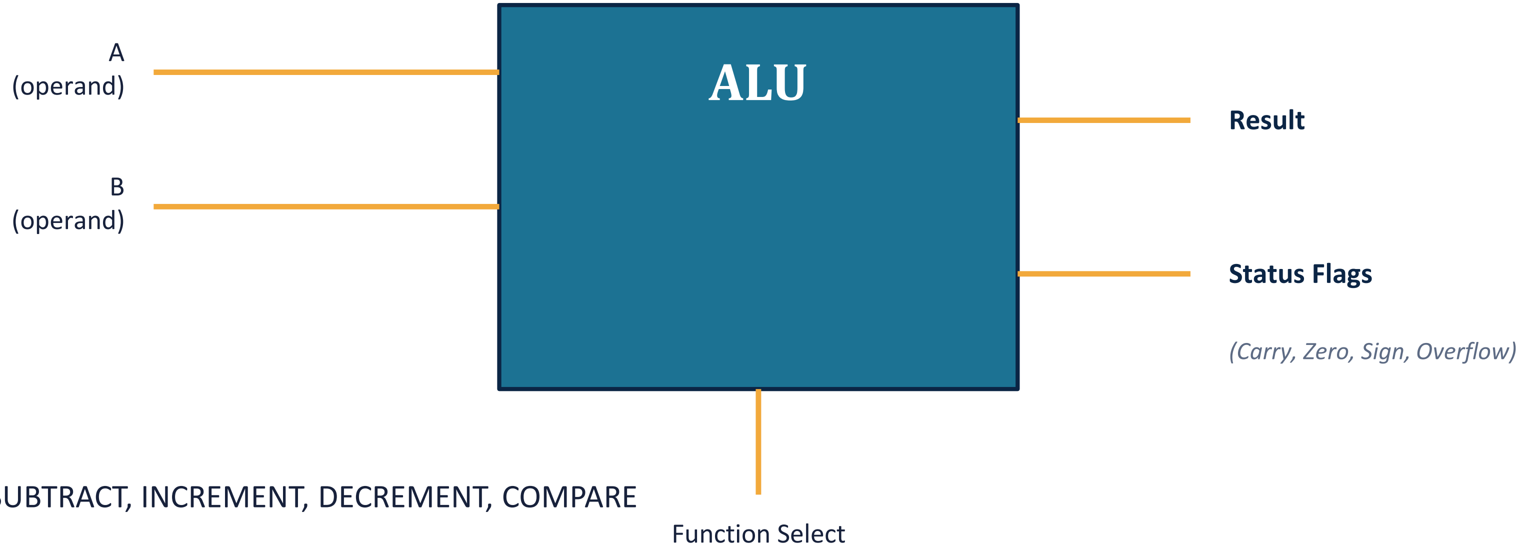
$$\text{Bout} = A'B + A'\text{Bin} + B\text{Bin}$$

PREFERRED METHOD — 2's COMPLEMENT SUBTRACTION

- In practice, digital systems avoid separate subtractor circuits entirely
- A – B is computed as A + (2's complement of B), reusing the same adder hardware
- 2's complement of B = (invert all bits of B) + 1
- Final carry-out (if any) is discarded — the adder result directly gives A – B

Arithmetic Logic Unit (ALU)

The ALU is a combinational circuit that performs a set of arithmetic and logic operations on binary data, selected by a function-select input — the computational core of a processor's datapath.



- Arithmetic group: ADD, SUBTRACT, INCREMENT, DECREMENT, COMPARE
- Logic group: AND, OR, XOR, NOT, shift / rotate operations
- A function-select code (e.g. 4 bits for 16 operations, as in the 74181) chooses the active operation

Representative ALU Operations

Select	Operation	Function
0000	ADD	R = A + B
0001	SUBTRACT	R = A – B
0010	INCREMENT	R = A + 1
0011	DECREMENT	R = A – 1
0100	AND	R = A · B
0101	OR	R = A + B (logic)
0110	XOR	$R = A \oplus B$
0111	NOT	R = A'
1000	SHIFT LEFT	R = A « 1
1001	SHIFT RIGHT	R = A » 1

STATUS FLAGS

Carry (C)

Set when an arithmetic operation produces a carry/borrow out of the MSB

Zero (Z)

Set when the result is exactly zero

Sign (S)

Reflects the MSB of the result (0 = positive, 1 = negative)

Overflow (V)

Set when a signed arithmetic result exceeds the representable range

Digital Comparators

A digital (magnitude) comparator compares two binary numbers, A and B, and indicates their relationship via three outputs: A > B, A = B, and A < B.

A	B	A>B	A=B	A<B
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

1-BIT EQUATIONS

$A>B = A \cdot B'$ $\quad$ $A<B = A' \cdot B$ $\quad$ $A=B = (A \odot B) = A'B' + AB$

The A = B output is essentially an XNOR of the two bits — the equality detector.

MULTI-BIT / CASCADED COMPARATORS

- Multi-bit numbers are compared starting from the MSB downward, using cascading inputs
- IC 7485 is a classic 4-bit magnitude comparator with cascade inputs to chain multiple ICs for wider words

Priority Encoder

An encoder converts an active input line into a binary code. A priority encoder resolves the case where multiple inputs are active simultaneously by encoding only the highest-priority (typically highest-index) input.

D3	D2	D1	D0	Y1	Y0	V
0	0	0	0	X	X	0
0	0	0	1	0	0	1
0	0	1	X	0	1	1
0	1	X	X	1	0	1
1	X	X	X	1	1	1

KEY IDEA

- X = don't care — lower-priority inputs are ignored once a higher one is active
- D3 has the highest priority; if D3 = 1, output = 11 regardless of D0–D2
- V (valid) flags whether any input is active at all
- Used in interrupt controllers to select the highest-priority request

Decoders

A decoder converts an n-bit binary code into one of 2^n unique active output lines — the reverse operation of an encoder.

A1	A0	Y0	Y1	Y2	Y3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

2-to-4 Line Decoder — exactly one output is HIGH (active) for each input code

APPLICATIONS

- Memory address decoding — selecting one memory chip/row from an address
- Instruction decoding in a CPU control unit
- Seven-segment display decoding (BCD → segment pattern)
- Building a demultiplexer (a decoder with a data-enable input)

BUILDING LARGER DECODERS

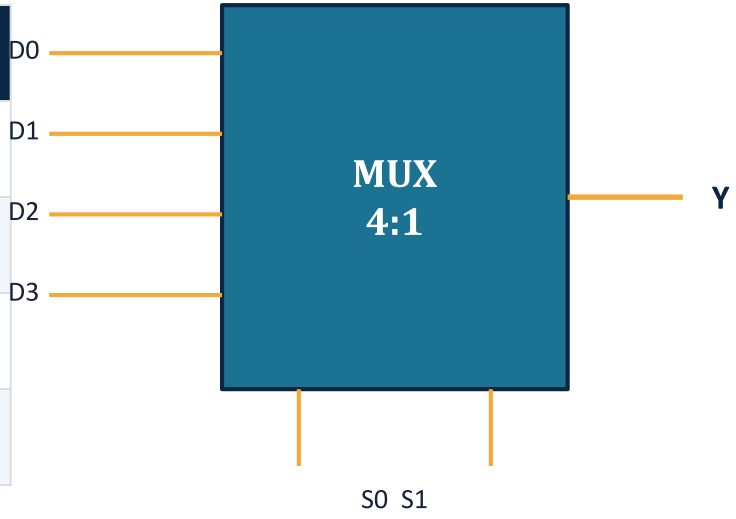
A 3-to-8 decoder can be built from two 2-to-4 decoders: the MSB of the 3-bit input drives the enable of each 2-to-4 block, so only one block is active and produces the correct one-of-eight output.

Multiplexer (MUX)

A multiplexer selects one of several input data lines and routes it to a single output, based on the value of select (control) lines — a data selector.

S1	S0	Y
0	0	D0
0	1	D1
1	0	D2
1	1	D3

4-to-1 MUX: 4 data inputs (D0–D3), 2 select lines, 1 output



APPLICATIONS

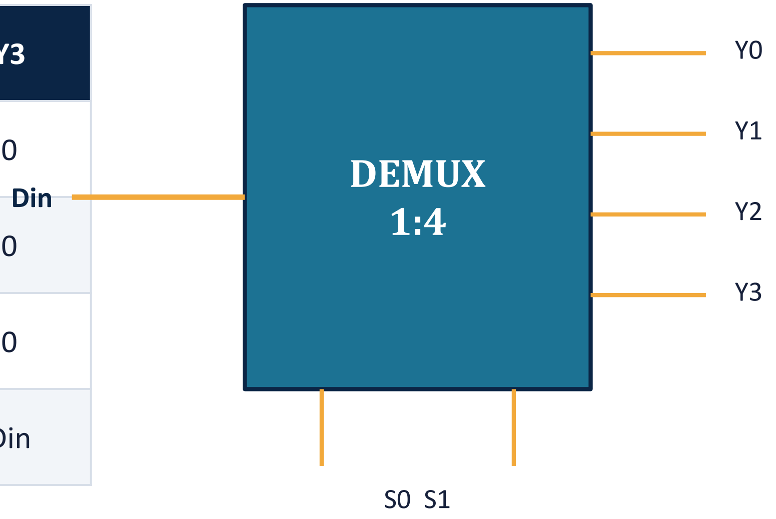
- Data routing on a shared bus
- Parallel-to-serial conversion
- Implementing Boolean functions directly from a truth table
- Selecting between multiple signal sources (e.g. sensors)

Demultiplexer (DEMUX)

A demultiplexer takes a single data input and routes it to one of several outputs, chosen by select (control) lines — the reverse of a multiplexer.

S1	S0	Y0	Y1	Y2	Y3
0	0	Din	0	0	0
0	1	0	Din	0	0
1	0	0	0	Din	0
1	1	0	0	0	Din

1-to-4 DEMUX: 1 data input (Din), 2 select lines, 4 outputs



APPLICATIONS

- Serial-to-parallel conversion
- Routing one data source to many destinations
- Memory write-decoding
- Combined with a decoder for enable-based addressing

SUMMARY

Key Takeaways

- Half adders combine two bits; full adders add a carry-in, enabling ripple-carry addition of multi-bit numbers.
- Binary subtraction is normally done via 2's complement addition, reusing adder hardware.
- The ALU packages arithmetic and logic operations behind a function-select code, with status flags reporting the result.
- Comparators, priority encoders, decoders, and MUX/DEMUX circuits handle data comparison, encoding, and routing — core building blocks of digital systems.

Thank You